Thought AI README

Value Proposition

The value proposition of ThoughtAI is to help students learn through guided, step-by-step problem-solving that reinforces critical thinking instead of shortcuts. By integrating AI directly into handwritten workflows, ThoughtAI keeps students in control of their reasoning, builds confidence in their understanding, and reframes AI as a learning partner rather than a replacement for human thought.

Access & Setup

You can access the Chatatouille Medium-Fi prototype through Figma using this link.

Once you follow the link, the prototype should already be set up and ready for use.

Intended Device

We chose the iPad scratch pad analyzer because it fits how students already study and requires no extra hardware. It supports natural handwritten problem solving and provides instant, contextual feedback directly on their work. Overall, it offers the best balance of accessibility, usability, and learning effectiveness.

Context

When is the app used?

- When students are actively solving homework problems (especially math or step-based subjects).
- During independent study sessions, they want feedback but don't have immediate access to a tutor.
- While reviewing mistakes to understand reasoning, not just the final answers.
- In real-time problem solving, immediate guidance supports focus and momentum.

What should a user be able to accomplish?

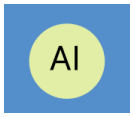
- Write or import a problem naturally (handwritten or photo).
- Receive feedback on correctness after submitting work.
- Enter Tutor Mode to be guided step-by-step through reasoning.
- Explain their thinking and unlock the next step only after demonstrating understanding.
- Learn from mistakes through hints instead of just being shown the solution.

Prototyping Tools

This prototype was built using Figma for interface design, wireframing, and interactive flow simulation. Some icons/buttons were sourced from Google, the rest were created within Figma. Basic click-through interactions were implemented using Figma's prototype linking features.

How to run/operate the prototype and accomplish the three task flows:

: Click to check existing work



: use to give next step

:Click to activate voice tutor and can be used to walk student through steps

: camera used to take pictures of problem the transfers them to notes app

:info page that provides instructions for the icons above

General run down of the app:

1. Start on the Welcome page:
    a. import button doesn't work
    b. Click new note -> click ok -> takes you to an empty note
    c. Click home page -> takes you to home page
2. When on homepage:

    a. Only notebook that works is Math, the rest is hard coded(English, Science, Add new)
    b. Click search button: takes you to search page
        i. Click math to go into math notebook
        ii. Click back button to go back to home page
    c. Each time, you exit out to home page:
        i. New note->takes you to empty notes page
        ii. Math-> takes you to page with basic math equation
3. From empty page:
    a. Click the back button to go to the home page
    b. Click camera icon -> click button how you normally take photos -> confirm by clicking ok or retake photo
    c. Once problem imported = how notebook looks when you click math from homepage
4. Homepage->math notebooks
    a. Simple task flow: Click into Math notebook
        i. :click this and the next portion of work will appear
            1. Click check icon and it will check the work(incorrect!)
            2. Click ok button
            3. Click eraser to go back to original problem
        ii. : once page back to original problem, click this and the next portion of work will appear
            1. Click check icon(correct!)
            2. Click ok to exit out
            3. Click eraser to get back to original problem
    b. Moderate task flow: Click AI button which will give the next step
        i. Click ok to exit out of this work flow
    c. Complex task flow: Click green voice button which will activate AI voice guide
        i. Exit at any time by clicking exit AI, can re-enter by clicking green voice button again
        ii. Keep clicking the voice memo to go through all the steps
        iii. Can access voice mode from the correct work flow

Limitations

The following limitations exist due to the nature of the Figma prototype:

- A real AI model does not actually power the AI feedback system. Responses are simulated.
- Handwriting recognition and problem analysis are not functional.
- The voice tutoring feature cannot process real speech input.
- The “Check Work” feature does not evaluate real mathematical correctness.
- Photo import does not actually scan or interpret written problems.
- Timing and system processing delays are simulated and do not reflect real AI response times.

These limitations were necessary because Figma does not support backend logic, AI processing, handwriting recognition, or real-time computation. The goal of this prototype was to test interaction flow, feedback structure, and user understanding rather than technical feasibility.

Wizard of Oz Techniques

To simulate AI intelligence and step-by-step tutoring, we used Wizard of Oz techniques:

- The AI tutor responses are pre-written and triggered through manual screen transitions.
- The “step-by-step” tutoring flow is scripted and does not adapt dynamically to user input.
- Correct and incorrect feedback states are manually designed rather than generated.
- The “Session Completed” summary screen is predetermined.

These techniques allowed us to evaluate whether guided reasoning improves understanding without building a real AI system.

Hard-Coded Items

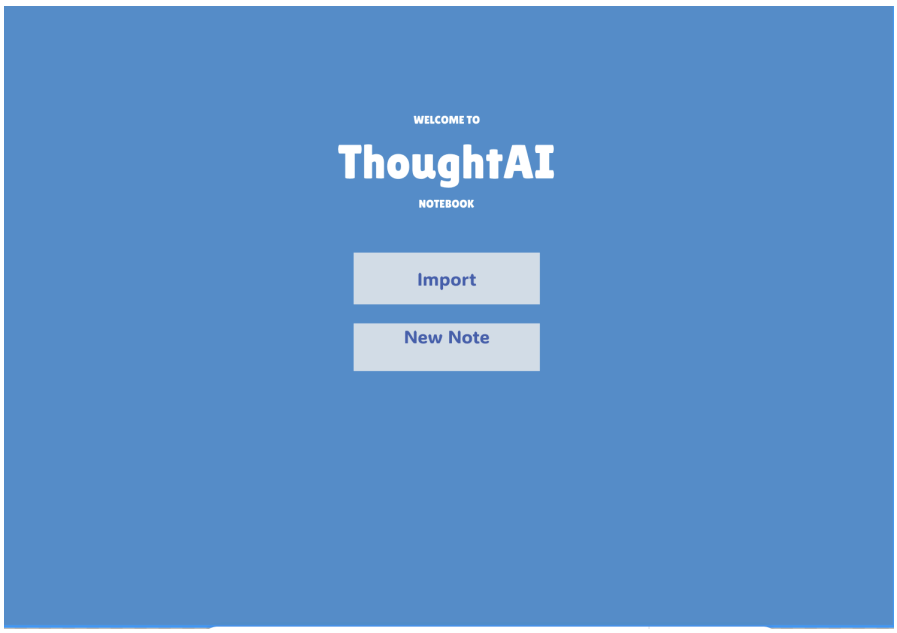
To overcome Figma’s lack of computational logic, several elements were hard-coded:

- The math problem shown (e.g., $3x + 5 = 20$) is fixed.

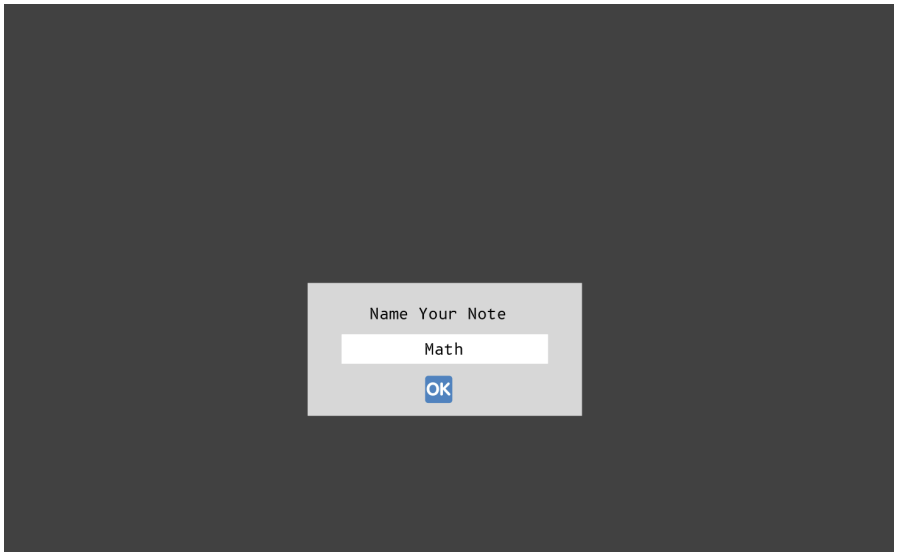
- The correct solution path is predetermined.
- Incorrect answer states always trigger the same hint.
- The Tutor Mode progression follows a fixed sequence of steps.
- The success message (“Session Completed”) is static.
- The AI prompt questions (e.g., “Why did you subtract 5 first?”) are pre-written and do not vary.

These hard-coded elements were necessary to simulate structured AI guidance while focusing on user experience, clarity of progression, and cognitive flow rather than technical implementation.

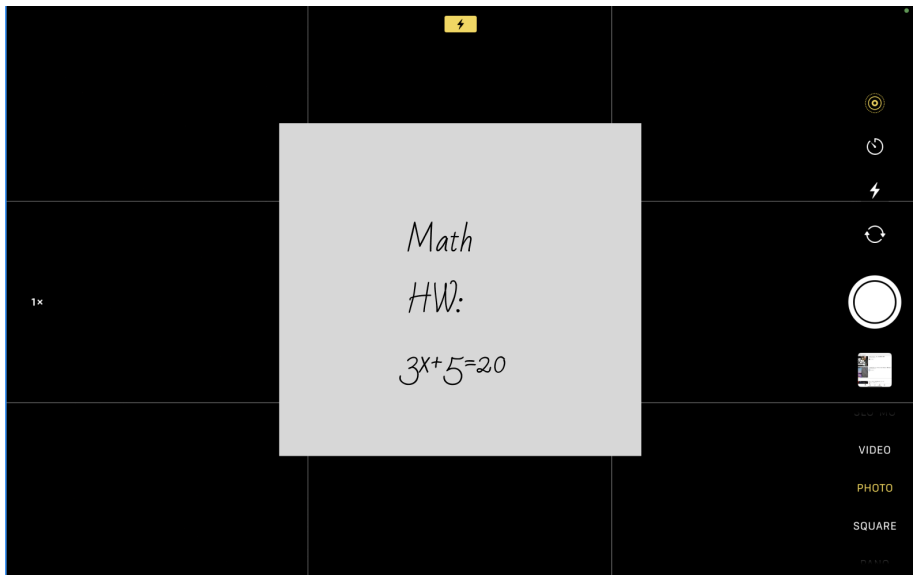
Images



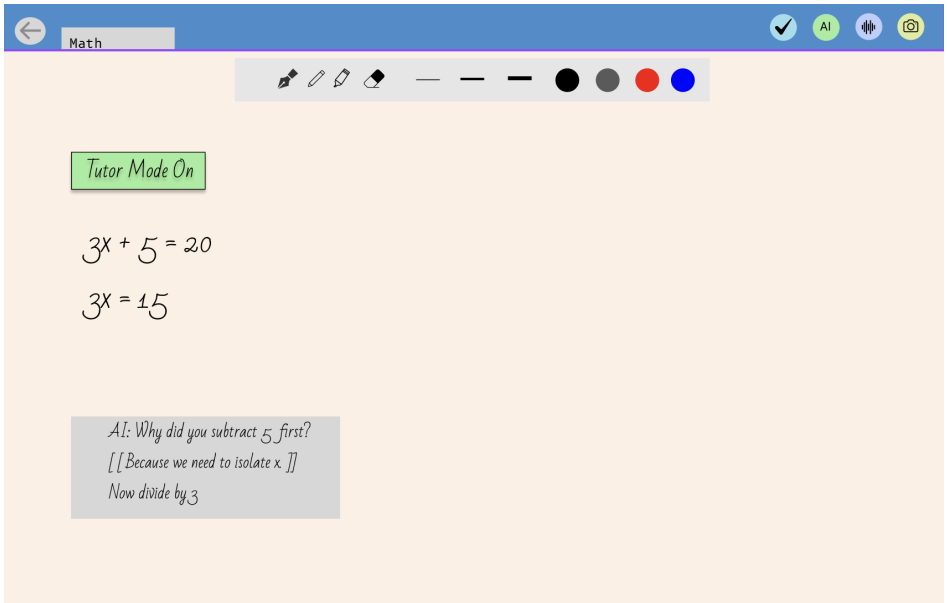
Home Page



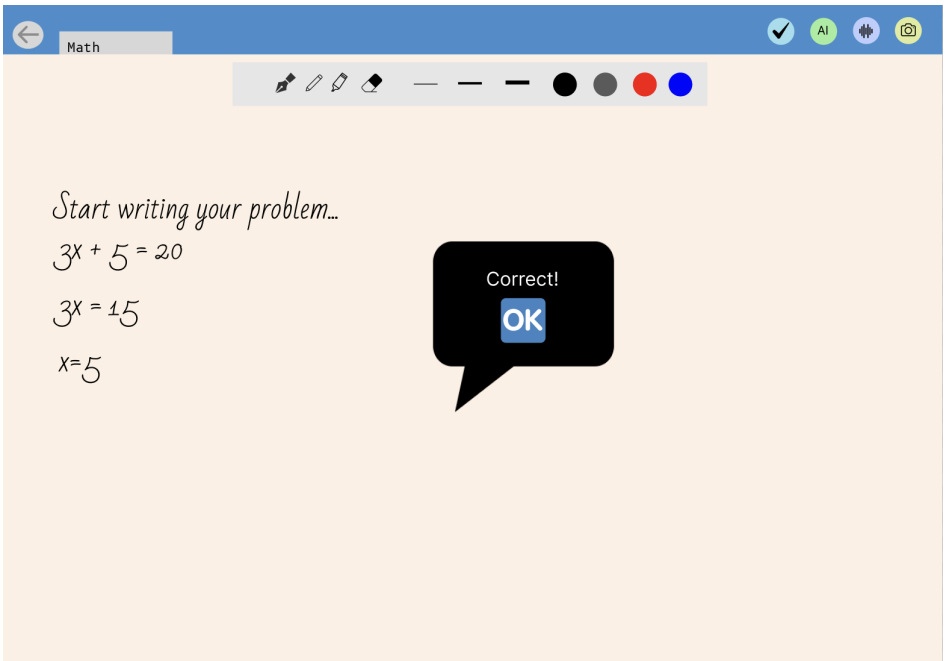
New note



Import



The user can write the solution on the board, and the AI helper will pop up at the bottom, as shown in the grey box



AI would grade the work after the user solved the problem